THE SURVIVAL PROBABILITY OF A HEAVY-TAILED CRITICAL DERRIDA–RETAUX SYSTEM

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ABSTRACT. For the Derrida–Retaux recursion $X_{n+1} = (X_n^{(1)} + \cdots + X_n^{(m)} - 1)^+$ in the critical regime $(m-1)\mathbb{E}(X_0 m^{X_0}) = \mathbb{E}(m^{X_0}) < \infty$, Chen, Derrida, Hu, Lifshits and Shi conjectured that $\mathbb{P}(X_n \neq 0) \sim 4(m-1)^{-2}n^{-2}$. We give a nonconstant critical law for $m = 2$, with $\mathbb{P}(X_0 = k) = c\,2^{-k}k^{-3}$ for $k \geq 2$, for which $\liminf_n n^2\,\mathbb{P}(X_n \neq 0) \leq 3/2$. So the conjecture does not hold under the critical moment condition alone. The bound $3/2$ is the constant that Chen, Hu and Shi predict for this tail. The proof combines an exact recursion for $\mathbb{E}(2^{X_n})$ with the upper bound of Chen and Shi on the product $\prod_{i<n}\mathbb{E}(2^{X_i})$.

1. INTRODUCTION

Fix an integer $m \geq 2$ and let X_0 be a random variable with values in $\mathbb{Z}_+ = \{0, 1, 2, \dots\}$. The Derrida–Retaux system is defined by

$$X_{n+1} \stackrel{d}{=} \big(X_n^{(1)} + \cdots + X_n^{(m)} - 1\big)^+, \qquad n \geq 0, \tag{1.1}$$

where $X_n^{(1)}, \dots, X_n^{(m)}$ are independent copies of X_n. It was introduced by Collet, Eckmann, Glaser and Martin [5] in a spin glass model, by Derrida, Hakim and Vannimenus [6] in a hierarchical model of wetting, and by Derrida and Retaux [7] as a toy model of the depinning transition. By [5], the critical regime is

$$(m-1)\,\mathbb{E}\big(X_0 m^{X_0}\big) = \mathbb{E}\big(m^{X_0}\big) < \infty. \tag{1.2}$$

Chen, Derrida, Hu, Lifshits and Shi [2] assume throughout that X_0 is not almost surely constant, prove that $\mathbb{E}(m^{X_n}) \to 1$ under (1.2) [2, Theorem 2], and state the following.

Conjecture 1 ([2]). *Assume* (1.2). *Then* $\mathbb{P}(X_n \neq 0) \sim \dfrac{4}{(m-1)^2}\,\dfrac{1}{n^2}$ *as* $n \to \infty$.

The same paper remarks that additional integrability, such as $\mathbb{E}(X_0^3 m^{X_0}) < \infty$, may be needed. Chen, Hu and Shi [3] proved $\mathbb{P}(X_n \geq 1) = n^{-2+o(1)}$ when $\mathbb{E}(t^{X_0}) < \infty$ for some $t > m$, and recorded the expectation that the constant $4/(m-1)^2$ requires $\mathbb{E}(X_0^3 m^{X_0}) < \infty$. For $m = 2$ and $\mathbb{P}(X_0 = k) \sim c_0 k^{-\alpha} 2^{-k}$ with $2 < \alpha \leq 4$ they expect instead $\mathbb{P}(X_n \geq 1) \sim \alpha(\alpha-2)/(2n^2)$ [3, (1.5)]; see also Derrida and Shi [8], and Chen, Dagard, Derrida and Shi [1] for the same range of critical behaviours in a coalescence equation. Hu, Mallein and Pain [9] give an exactly solvable continuous-time version of the model on the Yule tree and study its scaling limit at criticality. Our result proves that Conjecture 1 fails for such a law and gives the predicted constant as an upper bound for the lower limit.

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Let

$$A = \sum_{k\geq 2} \frac{2^{-k}}{k^3}, \qquad B = \sum_{k\geq 2} \frac{k-1}{k^3}, \qquad c = \frac{1}{A+B},$$

and let X_0 have the law

$$\mathbb{P}(X_0 = 0) = cB, \qquad \mathbb{P}(X_0 = 1) = 0, \qquad \mathbb{P}(X_0 = k) = \frac{c}{2^k k^3} \quad (k \geq 2). \tag{1.3}$$

Theorem 1.1. *Let $m = 2$ and let X_0 have the law* (1.3)*. Then X_0 is not almost surely constant,* (1.2) *holds,* $\mathbb{E}(X_0^3 2^{X_0}) = \infty$*, and*

$$\liminf_{n\to\infty} n^2\, \mathbb{P}(X_n \neq 0) \leq \frac{3}{2}.$$

In particular $\mathbb{P}(X_n \neq 0) \sim 4/n^2$ fails, and Conjecture 1 is false.

The constant law $X_0 \equiv 1$ with $m = 2$ also satisfies (1.2) and has $X_n \equiv 1$, but it is excluded by the standing assumption of [2]; the law (1.3) has $\mathbb{P}(X_0 = 0) > 0$ and $\mathbb{P}(X_0 = 2) > 0$.

The proof uses one external result, the upper half of a theorem of Chen and Shi.

Theorem 1.2 ([4, Theorem 1.1])**.** *Let X_0 satisfy* (1.2) *and $\mathbb{P}(X_0 \geq 2) > 0$, and suppose $\mathbb{P}(X_0 = k) \sim c_0 m^{-k} k^{-\alpha}$ as $k \to \infty$ for some $2 < \alpha < 4$ and $c_0 > 0$. Then there are constants $0 < c_1 \leq c_2$ such that for all $n \geq 1$,*

$$c_1 n^{\alpha-2} \leq \prod_{i=0}^{n-1} \mathbb{E}\big(m^{X_i}\big)^{m-1} \leq c_2 n^{\alpha-2}.$$

For the law (1.3) we have $\alpha = 3$, so $Q_n := \prod_{i<n} \mathbb{E}(2^{X_i}) \leq Cn$. Section 3 derives an exact recursion for $e_n = \mathbb{E}(2^{X_n}) - 1$, Section 4 shows that if $\mathbb{P}(X_n \neq 0)$ stays above a/n^2 with $a > 3/2$ then e_n stays above β/n for some $\beta > 1$, and Section 5 shows that this forces Q_n to grow faster than n.

2. The law

Lemma 2.1. *The numbers* (1.3) *define a probability law with $\mathbb{P}(X_0 = 0) > 0$ and $\mathbb{P}(X_0 = 2) > 0$. It satisfies*

$$\mathbb{E}(2^{X_0}) = \mathbb{E}(X_0 2^{X_0}) = c \sum_{k\geq 2} \frac{1}{k^2} < \infty,$$

which is (1.2) *for $m = 2$, and $\mathbb{E}(X_0^3 2^{X_0}) = \infty$.*

Proof. Both series defining A and B have positive terms and converge, since $2^{-k}k^{-3} \leq 2^{-k}$ and $(k-1)k^{-3} \leq k^{-2}$. Hence $c > 0$, all the weights are nonnegative, and they sum to $c(B + A) = 1$. Next,

$$\mathbb{E}(2^{X_0}) = cB + c\sum_{k\geq 2} \frac{1}{k^3}, \qquad \mathbb{E}(X_0 2^{X_0}) = c\sum_{k\geq 2} \frac{1}{k^2},$$

and these agree because $B = \sum_{k\geq 2}(k^{-2} - k^{-3})$. Finally $\mathbb{E}(X_0^3 2^{X_0}) = c\sum_{k\geq 2} 1 = \infty$. □

Since $\mathbb{P}(X_0 = k) = c\, 2^{-k} k^{-3}$ exactly for $k \geq 2$, Theorem 1.2 applies with $m = 2$, $\alpha = 3$ and $c_0 = c$, and gives a constant C with

$$Q_n = \prod_{i=0}^{n-1} \mathbb{E}(2^{X_i}) \leq Cn, \qquad n \geq 1. \tag{2.1}$$

3. An exact recursion

Write $G_n = \mathbb{E}(2^{X_n})$, $e_n = G_n - 1$ and $q_n = \mathbb{P}(X_n \neq 0)$. Each G_n is finite: $G_0 < \infty$ by Lemma 2.1, and $2^{(x+y-1)^+} \leq 2^{x+y}$ gives $G_{n+1} \leq G_n^2$. Since $2^x - 1 \geq \mathbf{1}_{\{x \geq 1\}}$ for $x \in \mathbb{Z}_+$,

$$0 \leq q_n \leq e_n. \tag{3.1}$$

Lemma 3.1. *For every $n \geq 0$,*

$$e_{n+1} = e_n + \frac{e_n^2}{2} - q_n + \frac{q_n^2}{2}. \tag{3.2}$$

Proof. Let $S = X_n^{(1)} + X_n^{(2)}$. On $\{S \geq 1\}$ we have $2^{(S-1)^+} = 2^S/2$, and on $\{S = 0\}$ we have $2^{(S-1)^+} = 1 = 2^S/2 + 1/2$. Hence

$$G_{n+1} = \frac{\mathbb{E}(2^S)}{2} + \frac{\mathbb{P}(S=0)}{2} = \frac{G_n^2}{2} + \frac{(1-q_n)^2}{2}.$$

Subtracting 1 and writing $G_n = 1 + e_n$ gives (3.2). □

4. A barrier

Lemma 4.1. *Let $a > 0$ and $\beta \geq 0$ satisfy $\gamma := a - \beta - \beta^2/2 > 0$. Let (e_n) and (q_n) be nonnegative sequences satisfying (3.2) for all n, and suppose there is $N \geq 1$ with*

$$q_n - \frac{q_n^2}{2} \geq \frac{a}{n^2} \qquad \text{for all } n \geq N. \tag{4.1}$$

Then $e_n > \beta/n$ for all $n \geq N$.

Proof. Put $b_n = \beta/n$ and $d_n = e_n - b_n$. Let $n \geq N$ with $d_n \leq 0$. Then $0 \leq e_n \leq b_n$, so $e_n^2 \leq \beta^2/n^2$, and (3.2) with (4.1) gives

$$e_{n+1} \leq e_n + \frac{\beta^2}{2n^2} - \frac{a}{n^2}.$$

Since $b_n - b_{n+1} = \beta/(n(n+1)) \leq \beta/n^2$, it follows that

$$d_{n+1} \leq d_n - \frac{\gamma}{n^2}. \tag{4.2}$$

In particular $d_{n+1} < 0$. Suppose now that $d_j \leq 0$ for some $j \geq N$. By induction (4.2) applies at every $k \geq j$, the sequence $(d_k)_{k \geq j}$ decreases, and $d_k \leq -\gamma/j^2$ for all $k > j$. Then $e_k = b_k + d_k \leq \beta/k - \gamma/j^2 < 0$ for all large k, which contradicts $e_k \geq 0$. Hence $d_n > 0$ for every $n \geq N$. □

5. Proof of Theorem 1.1

The first three assertions are Lemma 2.1. Suppose, for a contradiction, that $L := \liminf_n n^2 q_n > 3/2$ (possibly $L = \infty$). Choose a with $3/2 < a < L$ and then $\beta > 1$ so close to 1 that $\beta + \beta^2/2 < a$. By [2, Theorem 2], $e_n \to 0$, so $q_n \to 0$ by (3.1). Fix a' with $a < a' < L$. For large n we have $n^2 q_n \geq a'$ and $1 - q_n/2 \geq a/a'$, so

$$q_n - \frac{q_n^2}{2} = q_n\Big(1 - \frac{q_n}{2}\Big) \geq \frac{a'}{n^2} \cdot \frac{a}{a'} = \frac{a}{n^2}.$$

Thus (4.1) holds from some N on, and Lemma 4.1 gives $e_i > \beta/i$ for all $i \geq N$.

For $x \ge 0$ we have $\log(1+x) \ge x - x^2/2$, as both sides vanish at 0 and the derivative of the difference is $x^2/(1+x) \ge 0$. Since $G_i \ge 1$ for every i, for $n > N$

$$\log Q_n \ge \sum_{i=N}^{n-1} \log\Big(1 + \frac{\beta}{i}\Big) \ge \beta \sum_{i=N}^{n-1} \frac{1}{i} - \frac{\beta^2}{2} \sum_{i \ge N} \frac{1}{i^2} \ge \beta \log \frac{n}{N} - \frac{\beta^2}{2} \sum_{i \ge N} \frac{1}{i^2}.$$

Hence $Q_n \ge K n^\beta$ for some $K > 0$ and all $n > N$. Together with (2.1) this gives $K n^{\beta-1} \le C$ for all large n, which is impossible because $\beta > 1$. Therefore $\liminf_n n^2 q_n \le 3/2 < 4$. □

Remark 5.1. Only the case $(a, \beta) = (3, 3/2)$ of Lemma 4.1 is needed to refute $q_n \sim 4/n^2$: then $q_n - q_n^2/2 \ge 3/n^2$ eventually, $e_n > 3/(2n)$, and $Q_n \ge K n^{3/2}$. The argument does not determine the asymptotics of q_n, and it says nothing about laws with $\mathbb{E}(X_0^3 m^{X_0}) < \infty$, for which Conjecture 1 remains open. For the law (1.3) the truncated moment $\mathbb{E}(X_0^3 2^{X_0} \mathbf{1}_{\{X_0 \le k\}})$ equals $c(k-1)$ for $k \ge 2$, so it grows linearly, in line with $Q_n \le Cn$.

6. Computations

The repository [10] contains a program and a Lean file. They are not part of the proof. The program `checks.py` evaluates A, B, c, $\mathbb{E}(2^{X_0})$ and $\mathbb{E}(X_0 2^{X_0})$ to 50 digits through $A = \mathrm{Li}_3(1/2) - 1/2$, $B = \zeta(2) - \zeta(3)$, $\mathbb{E}(2^{X_0}) = c(B + \zeta(3) - 1)$ and $\mathbb{E}(X_0 2^{X_0}) = c(\zeta(2) - 1)$; to 15 digits, $c = 2.08294123137751$, $\mathbb{P}(X_0 = 0) = 0.922487104682579$, and both moments equal 1.34335975935815. In exact rational arithmetic it checks the identity behind (3.2) and the one-step inequality (4.2) for $(a, \beta) = (3, 3/2)$ and $(7/4, 11/10)$, at $n = 3, \dots, 202$ and 21 values of e_n in $[0, b_n]$ each, with $q_n = a/n^2 + a^2/n^4$ and $q_n = 2a/n^2$ (both satisfy (4.1)), and the truncated moment identity of the remark for $k \le 600$. The Lean file `Barrier.lean` proves the polynomial identity behind (3.2) and the inequality (4.2) for $(a, \beta) = (3, 3/2)$, with Lean 4 and Mathlib.

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This work was done by multiple AI agents, Claude (Anthropic) and Codex (OpenAI), connected to Math Brain [11] and working together inside the Brain framework [13], a shared, persistent memory in which the agents record claims, proofs, failed attempts and methods, retrieve one another's results, coordinate who works on what, and audit each other's work. For more on the details behind the Brain and Math Brain, see the Math Brain paper [12], the Math Brain repository [11] and the Brain framework repository [13].

Declaration of competing interest

The author holds equity in Heights Labs and in Convex Nexus Capital and has a management role at Convex Nexus Capital. These entities are included in the funding declaration above.

Declaration of generative AI and AI-assisted technologies

The paper comes out of Math Brain, an ongoing project on the use of AI systems for research mathematics. In October 2026 OpenAI GPT models, running as research tasks of that project, found the law (1.3), proved that it does not satisfy $\mathbb{P}(X_n \neq 0) \sim 4/n^2$ by the argument of the remark in Section 5, and wrote the Lean file. In October and November 2026 Claude Opus 5.5 (Anthropic), working in Claude Code, checked the proof, extended it to the bound $\liminf_n n^2\mathbb{P}(X_n \neq 0) \leq 3/2$, wrote the verification program and wrote the paper. The author directed the work and is responsible for the content of the paper and for decisions concerning its dissemination.

Data availability

No data were used. The code is available at `https://github.com/yspennstate/derrida-retaux-critical-survival`.

The paper is part of the Math Brain project, whose public repository is `https://github.com/yspennstate/math-brain-public` [11].

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